Semi-Blind Spectral Deconvolution with Adaptive Tikhonov Regularization

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Deconvolution has become one of the most used methods for improving spectral resolution. Deconvolution is an ill-posed problem, especially when the point spread function (PSF) is unknown. Non-blind deconvolution methods use a predefined PSF, but in practice the PSF is not known exactly. Blind deconvolution methods estimate the PSF and spectrum simultaneously from the observed spectra, which become even more difficult in the presence of strong noise. In this paper, we present a semi-blind deconvolution method to improve the spectral resolution that does not assume a known PSF but models it as a parametric function in combination with the a priori knowledge about the characteristics of the instrumental response. First, we construct the energy functional, including Tikhonov regularization terms for both the spectrum and the parametric PSF. Moreover, an adaptive weighting term is devised in terms of the magnitude of the first derivative of spectral data to adjust the Tikhonov regularization for the spectrum. Then we minimize the energy functional to obtain the spectrum and the parameters of the PSF. We also discuss how to select the regularization parameters. Comparative results with other deconvolution methods on simulated degraded spectra, as well as on experimental infrared spectra, are presented.

Index Headings: **Infrared spectroscopy; Spectral deconvolution; Semi-blind deconvolution; Tikhonov regularization.**

INTRODUCTION

Deconvolution is a mathematical technique that removes the broadening effect of the instrumental response function (or point spread function, PSF) to improve the spectral resolution. Many spectral deconvolution methods have been developed. There are two major types of methods: the non-blind deconvolution (NBD) method, in which the PSF is assumed to be known in advance, and the blind deconvolution (BD) method, in which the PSF is unknown.

Non-blind deconvolution methods have been widely researched for spectral deconvolution. Wiener filtering is the most popular.[1] Fourier self-deconvolution (FSD), developed by Kauppinen et al., is the most common method used in infrared spectroscopy.[2] The constrained deconvolution algorithm with a relaxation weighting function proposed by Jansson has been proved to be practically effective.[3] Other methods, such as the maximum likelihood, maximum entropy, and alternating projection methods, have also achieved a certain amount of success.[4,5] The maximum Burg entropy (MaxEntD) method suggested by Lórenz-Fonfría has obtained impressive narrowing and noise suppression.[5] In practice, however, the instrumental response function is not known exactly. In addition to the physical mechanism and the design scheme, the adjustment and aging of the instrument can also affect its response function. Also, NBD methods are very sensitive to the mismatch between the used and the true PSF, and a poor knowledge of the PSF normally leads to a poor recovery result.

In contrast, without assuming a known PSF, blind deconvolution methods estimate the PSF and the spectrum simultaneously from the measured data. Senga et al. developed a homomorphic filtering method and applied it to the infrared molecular absorption spectra successfully.[6] Zou et al. proposed an iterative method, starting from an initial fitting response function, to iteratively estimate the true spectrum and upgrade the response function.[7] Yuan et al. proposed a high-order statistical Gauss–Newton algorithm to blindly deconvolve the measured spectroscopic data.[8] Blind deconvolution becomes even more difficult because the PSF is not known in advance, and the presence of noise will further deteriorate the problem.[7,8]

Because the PSF is very important for deconvolution, we should adequately utilize the a priori knowledge about the instrumental PSF to improve deconvolution performance. In the case of non-blind deconvolution methods, assuming a known PSF is irrational, but for blind deconvolution methods, not utilizing any information about the PSF is also irrational. In fact, many studies have shown that the overall instrumental response function approaches a Gaussian form, which is especially true if the factors of broadening are multifold, such as slit function, grating response, circuit response, etc.[3,7] Therefore, the PSF can be modeled by a parametric function and applied in the deconvolution process, for example, a Gaussian function, which is completely defined by a single parameter (the Gaussian's variance). Borrowing from a similar concept in image restoration, we refer to this as the semi-blind spectral deconvolution (SBD) method in this paper. It is expected that the SBD method will produce better results because the number of unknowns in the PSF is reduced from

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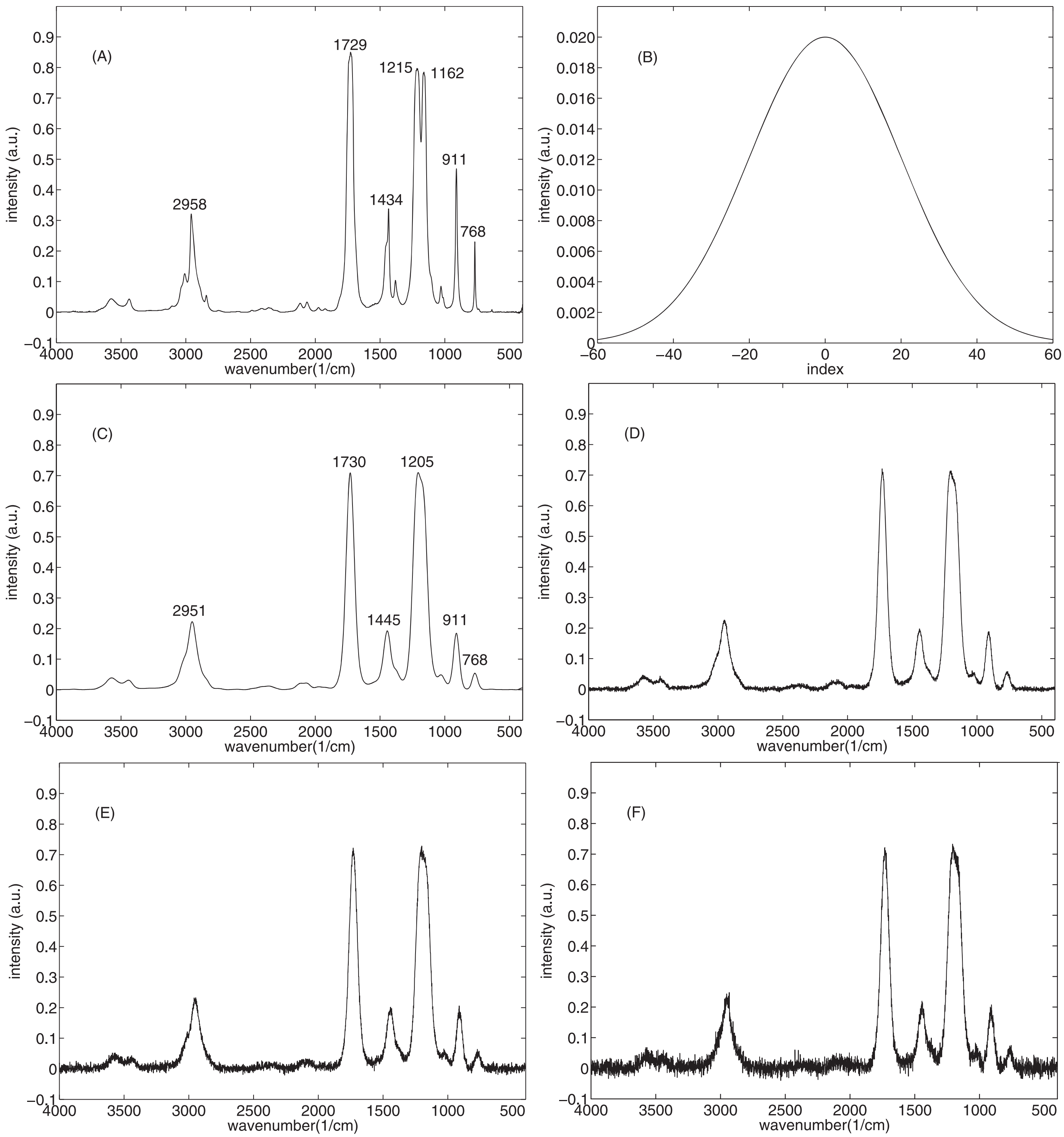

FIG. 1. Simulated spectrum. (**A**) Original infrared spectrum. (**B**) Gaussian kernel with $\sigma = 20$. (**C**) Convolution between (A) and (B). Noisy spectrum with (**D**) SNR $= 200$. (**E**) SNR $= 100$. (**F**) SNR $= 50$.

tens or hundreds of data to a small number of parameters that describe the PSF. Experimental results have proved its effectiveness. To our knowledge, we introduce semi-blind methods for spectral deconvolution for the first time.

All deconvolution methods suffer the ill-posed problem especially in the presence of noise in the measured spectra, even if the noise is very weak.[9–11] The problem becomes more serious as the noise increases. To settle the problem, most methods incorporate the a priori knowledge about the spectra into a regularization functional to constrain the obtained solution.[4,5,12] In practice, what and how the a priori knowledge can be used are the main questions. Mainly the assumption of solution smoothness is used.[4,12] Tikhonov regularization is commonly used to constrain a smooth solution.[4,5,13–15] Maximum entropy deconvolution methods often assume the non-negativity of the spectral data, Lórenz-Fonfría's method

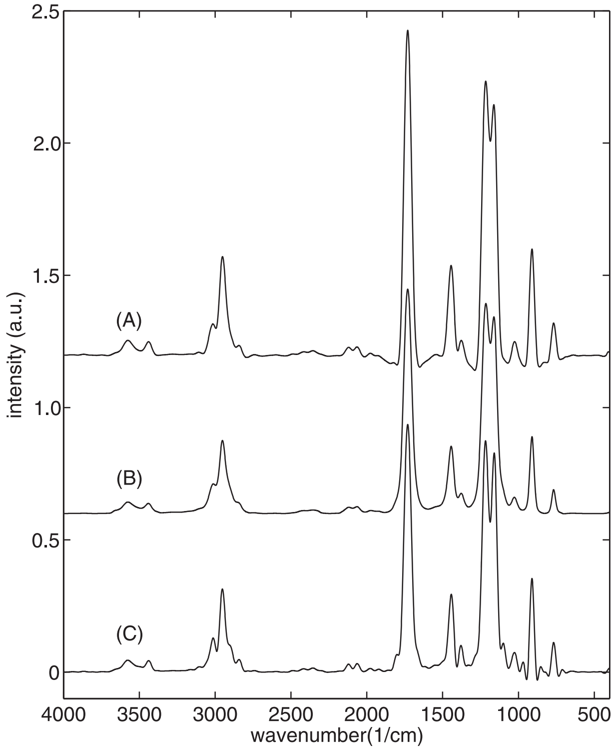

FIG. 2. Deconvolution results of the noise-free spectrum (see Fig. 1C). (**A**) FSD method (narrowing factor is 2.2). (**B**) MaxEntD method (NEF = 2). (**C**) SBD method ($\alpha = 0.05$, $\beta = 300$).

uses Burg entropy as the regularization functional without sign restriction, thus permitting negative values for the solution.[5]

Indeed, these conventional a priori constraints have generic and global characteristics, leading to the imposition of holistic constraints on the whole spectrum while reducing or even losing the local structures of the recovery spectrum. For example, the smoothness constraints often favor a smooth solution in order to reject a noisy solution, but at the same time, the constraint makes the abrupt peaks more flat and rejects the sharp transitions in the spectral signal. However, those local details sometimes relate to the intense absorption bands, which are very important for the interpretation of the spectrum. In order to preserve the local details, some local information should be further extracted from the spectrum being processed to adaptively adjust regularization. How to extract and formulate the spectrum-derived information deserves further study. The way that we propose to accomplish this end is to weight the regularization based on the magnitude of the first derivative of each spectral data point. With this approach, the regularization is able to adaptively preserve detail, thus outperforming the conventional regularized methods.

In this paper, we proposed a semi-blind spectral deconvolution method with adaptive Tikhonov regularization. We modeled the instrumental response function in parametric form and then converted PSF estimation into parameter optimization. The regularization of the spectrum was adaptively weighted in terms of the magnitude of the first derivative of each spectral data point. The proposed method does not need a known PSF in advance and has the capability of preserving detail. We tested and compared the performance of the proposed method with FSD and MaxEntD deconvolution methods on simulated infrared spectra. Deconvolution was also tested on infrared experimental data.

THEORY AND METHOD

Spectral Deconvolution Problem. The measured signal at the output of the spectral instrument $g(v)$ can be represented as the linear convolution of a more resolved signal $f(v)$ with the point spread function (PSF) $h(v)$, as well as a random process $n(v)$ superposed on the useful signal, which describes random errors arising during measurement of the spectrum, i.e.,

$$g(v) = f(v) \otimes h(v) + n(v) \tag{1}$$

where $\otimes$ denotes the linear convolution operation and is defined by the integral $f(v) \otimes h(v) = \int f(z)h(v - z)\, \mathrm{d}z$. The PSF $h(v)$ stands for the overall instrumental response function (also called the blur kernel), which collects the intrinsic line-shape function and the instrumental broadening.

The deconvolution problem, as applied to spectroscopic data, can be formulated as follows: given the measured data $g(v)$, with the help of the available knowledge about the PSF $h(v)$ and statistical characteristics of random noise $n(v)$, Eq. 1 can be solved to estimate $f(v)$, consequently increasing

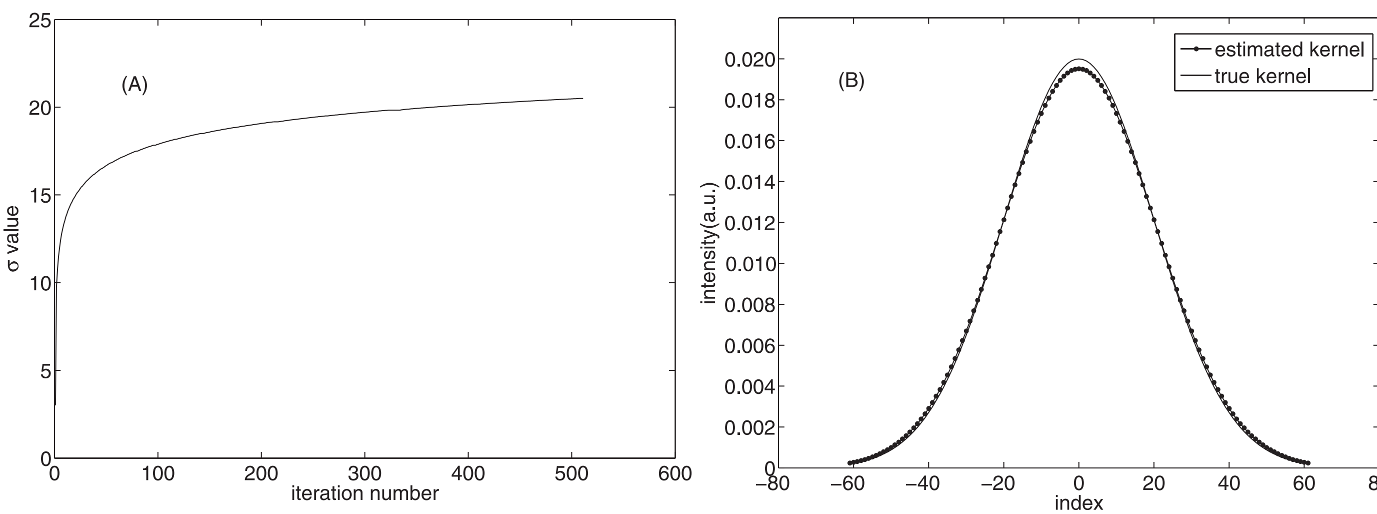

FIG. 3. Blur kernel estimated using SBD method on the noise-free spectrum (see Fig. 1C). (**A**) Convergence curve of the estimated width σ. (**B**) Estimated kernel.

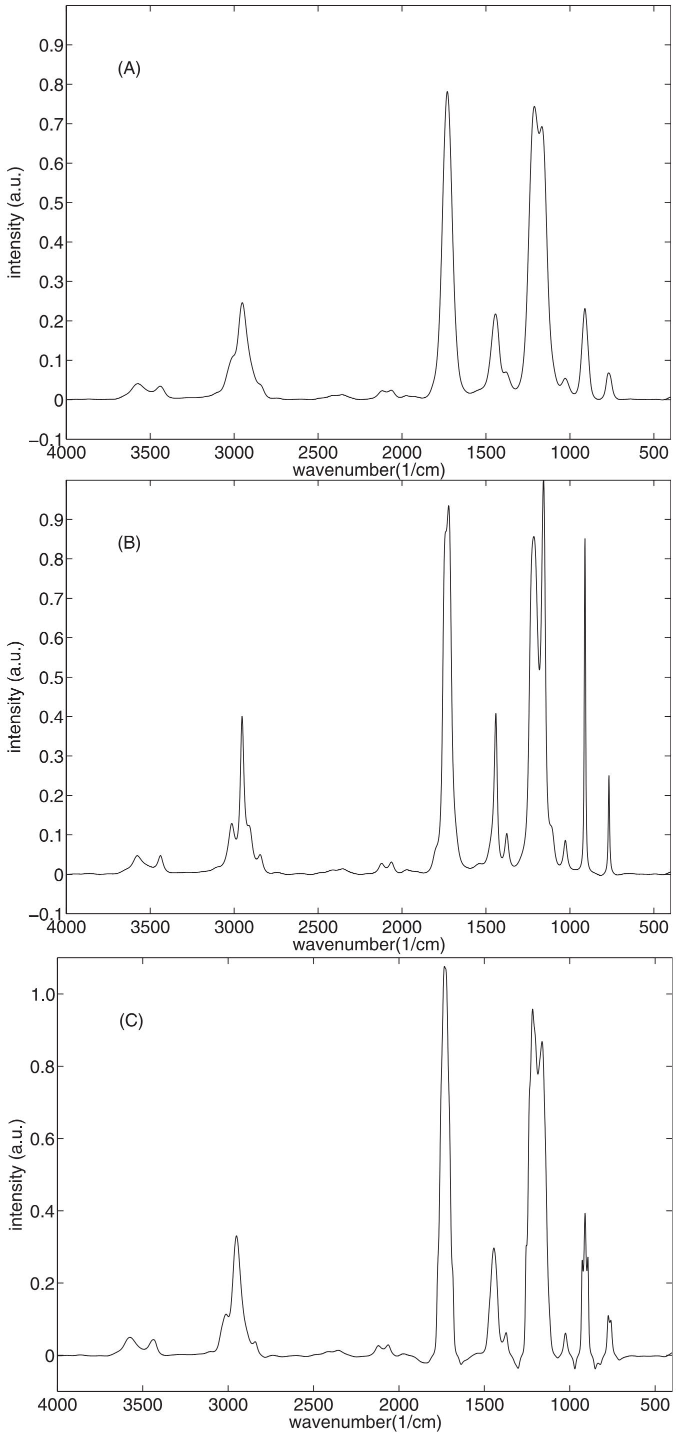

FIG. 4. Deconvolution results using MaxEntD method with inaccurate kernels. (**A**) Gaussian kernel with $\sigma = 15$. (**B**) Gaussian kernel with $\sigma = 25$. (**C**) Lorentzian kernel with $\gamma = 15$.

resolution. It is well known that, in the general case, such a problem is ill-posed. The blur kernel $h(v)$ is important for deconvolution. In the case of non-blind deconvolution methods, $h(v)$ is assumed to be known exactly, while in the blind-deconvolution methods $h(v)$ is completely unknown but estimated along with $f(v)$ from the measured data. In this paper, $h(v)$ is modeled as a parametric form in terms of the a priori knowledge about the characteristics of the instrumental response. With the parametric kernel, the blind deconvolution problem is reduced to a semi-blind deconvolution problem, which can help produce better results.

Regularized Semi-Blind Deconvolution. Deconvolution of a blurred and noisy spectrum is difficult even if the blur kernel $h(v)$ is known. Formally, finding $f(v)$ that minimizes

$$E(f) = \frac{1}{2}\int_L (f \otimes h - g)^2 \, \mathrm{d}v \tag{2}$$

is an ill-posed inverse problem: small perturbations in the data may produce unbounded variations in the solution (where L denotes the integral interval defined on the spectral wavenumber range).[4,5] To settle the problem, some a priori knowledge about the solution is applied to obtain a regularized solution. In Tikhonov regularization,[13–17] smoothness about the solution is imposed and measured by the L_2 norm of its first derivative, and the energy functional takes the following form:

$$E(f) = \frac{1}{2}\int_L (f \otimes h - g)^2 \, \mathrm{d}v + \alpha \int_L |f'|^2 \, \mathrm{d}v \tag{3}$$

where f' denotes the first derivative of f with respect to variable v, and $|f'|$ is the absolute value of f'. The parameter α controls the smoothness of the recovered spectrum. As α is increased, the recovered spectrum becomes smoother, at the expense of lower data fidelity and a narrow kernel.

The deconvolution becomes even more difficult if the blur kernel is not known in advance. In addition to being ill-posed with respect to the spectrum, the blind deconvolution problem is ill-posed in the kernel as well.[4,10,18] In our semi-blind deconvolution method, the kernel is defined by the parametric function h_ξ, in which the subscript ξ is the unknown parameter. In this work, the kernel is modeled as a Gaussian function h_σ by its standard deviation σ. As a result, the blind deconvolution problem is reduced to the semi-blind deconvolution, which helps alleviate the ill-posed problem. Furthermore, a Tikhonov regularization term about the kernel $\int_L |h'_\sigma|^2 \, \mathrm{d}v$ is added to the functional in Eq. 3. The energy functional about the spectrum f and the kernel parameter σ is thus:[18–20]

$$E(f,\sigma) = \frac{1}{2}\int_L (f \otimes h - g)^2 \, \mathrm{d}v + \alpha \int_L |f'|^2 \, \mathrm{d}v + \beta \int_L |h'_\sigma|^2 \, \mathrm{d}v \tag{4}$$

where h'_σ denotes the first derivative of h_σ with respect to v, and α and β are the regularization parameters that control the smoothness of the recovered spectrum and the kernel, respectively.

Tikhonov regularization often leads to over-smoothing in solutions and loss of sharp information.[18,19] For the kernel, the assumption of smoothness is rational because the response function is often smooth. But for the spectrum, over-smoothing constraints will lead to the reduction or even loss of intense and sharp band information in the spectra. However, these spectral details are very important in the interpretation of the spectra. In order to better preserve detail information, in the Tikhonov regularization term we added an adaptive weighting function $\rho(|f'|)$ to control the regularization about the spectrum:

$$E(f,\sigma) = \frac{1}{2}\int_L (f \otimes h_\sigma - g)^2 \, \mathrm{d}v + \alpha \int_L \rho(|f'|)|f'|^2 \, \mathrm{d}v + \beta \int_L |h'_\sigma|^2 \mathrm{d}v \tag{5}$$

TABLE I. **Band distortions in deconvoluted spectra by MaxEntD and SBD method.**

Band position		2958	1729	1434	1215	1162	911	768	RMSE
Position	MaxEntD	−4	0	+6	−2	−1	0	−1	2.879
	SBD	−6	+1	+9	+2	−2	−1	0	4.259
FWHM	MaxEntD	+14	+33	+10	–[a]	–	+18	+5	18.623
	SBD	+6	+16	+1	–	–	+16	+21	14.071
Height	MaxEntD	−0.041	−0.009	−0.080	−0.009	−0.040	−0.329	−0.083	0.134
	SBD	−0.007	+0.086	−0.044	+0.078	+0.043	−0.116	−0.118	0.080
Area[b]	MaxEntD	+2.069	+27.498	−0.357	–	–	−2.747	0.078	10.475
	SBD	+1.564	+17.470	−1.330	–	–	+3.808	+1.405	6.821

[a] FWHM and area cannot be calculated exactly for overlapped bands.
[b] Area is calculated as the product of FWHM and height here."+" or "−"indicates larger or smaller than the original, respectively.

where $\rho(|f'|)$ takes the form

$$\rho(|f'|) = \exp[-(|f'|/k)^2] \tag{6}$$

and k is a constant that determines the decreasing speed of the weighting function. The smaller k is, the faster the weighting function decreases. In the spectral region containing abrupt and intense bands, $|f'|$ takes on larger values, and thus the weighting function will take on smaller values, which places a smaller smoothing penalty on these regions containing detail, therefore preserving the detail information. On the other hand, in the flat spectral region, $|f'|$ takes on smaller values, leading to larger weighting of the function values, which will impose a larger penalty on these flat regions, thus strengthening smoothing of the noise. Therefore, the proposed semi-blind deconvolution method can adapt the regularization to the spectrum.

Optimization and Numerical Solution. The functional given in Eq. 5 depends on the spectrum f and the width parameter σ of the blur kernel h_σ. The proposed semi-blind deconvolution method is to find the optimal f and σ that minimize the energy functional (Eq. 5). Minimization with respect to f is carried out using the Euler–Lagrange (E-L) equation (Eq. 7), with Neumann boundary conditions. The E-L equation is a linear partial differential equation. Minimization with respect to the parameter σ is determined by differentiation of the objective functional in Eq. 8. The mathematical derivations are presented in the Appendix.

$$\begin{aligned}\frac{\delta E}{\delta F} &= (f \otimes h_\sigma - g) \otimes h_\sigma(-v) \\ &\quad - 2\alpha \exp\left(-\frac{f'^2}{k^2}\right) f''\left(\frac{2f'^4}{k^4} - \frac{5f'^2}{k^2} + 1\right) \\ &= 0\end{aligned} \tag{7}$$

$$\frac{\partial E}{\partial \sigma} = \int_L \left[(h_\sigma \otimes f - g)\left(\frac{\partial h_\sigma}{\partial \sigma} \otimes f\right) + \beta \frac{\partial}{\partial \sigma}|h'_\sigma|^2\right] \mathrm{d}v = 0 \tag{8}$$

In Eq. 7 f'' is the second derivative of f with respect to v, and in Eq. 8 below,

$$h_\sigma(v) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{v^2}{2\sigma^2}\right) \tag{9}$$

$$\frac{\partial h_\sigma}{\partial \sigma} = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{v^2}{2\sigma^2}\right) \cdot \left(\frac{v^2}{\sigma^2} - 1\right) \tag{10}$$

$$\frac{\partial}{\partial \sigma}|h'_\sigma|^2 = \frac{v^2}{\pi\sigma^7} \exp\left(-\frac{v^2}{\sigma^2}\right)\left(\frac{v^2}{\sigma^2} - 3\right) \tag{11}$$

The iterative alternate minimization (AM) approach[18,19,21] is applied: at the beginning, the recovered spectrum f is initialized as the observed spectrum g and σ as the given σ_0, and then in each step of the iterative procedure we minimize with respect to one function and keep another fixed. The procedure is illustrated as follows:

Initialization: $f = g$, $\sigma = \sigma_0$, $\sigma_k \gg 1$
while ($|\sigma_k - \sigma| > \varepsilon$) repeat
1. Solve Eq. 7 for f using gradient descent flow method.
2. Set $\sigma_k = \sigma$, and solve Eq. 8 for σ using bisection method.

Here ε is a small positive constant between 10^{-7} and 10^{-9}. The essence of the proposed method is iterative solution of the linear partial differential equations 7 and 8. They are discretized and solved numerically. The derivatives are discretized by finite differences. The first derivative f' is approximated by central finite differences $\Delta f_i = (f_{i+1} - f_i)/2$, and the second derivative f'' is approximated by the forward and backward finite difference $\Delta_+(\Delta_- f_i) = f_{i+1} + f_{i-1} - 2f_i$. The subscript i denotes the ith data point of the spectrum.

In step 1, the gradient descent flow method[10,11,18] is used for solving Eq. 7. Following discretization, the desired solution is obtained by employing the successive approximations iteration:

$$f^{n+1} = f^n + \Delta t\left(-\frac{\delta E}{\delta f}\right) \tag{12}$$

where the superscript n is the iteration number and Δt is the step size. If Δt is too small, the convergence will be very slow. On the other hand, if it is too large, the algorithm will be unstable or divergent. The typical values for Δt used in this paper were between 0.5 and 0.8.

In step 2, Eq. 8 takes the discrete form

$$\sum_{i \in L}(h_\sigma \otimes f - g)\left(\frac{\partial h_\sigma}{\partial \sigma} \otimes f\right)_i + \beta \sum_{i \in L}\left(\frac{\partial}{\partial \sigma}|h'_\sigma|^2\right)_i = 0 \tag{13}$$

where $\partial h_\sigma/\partial\sigma$ and $\partial|h'_\sigma|^2/\partial\sigma$ are given in Eqs. 10 and 11, respectively. The equation is solved for σ using the bisection method.[22] The discrete support of the Gaussian kernel was limited to $6\sigma + 1$, which in our experiments was much smaller than the spectrum size. The numerical integral of h_σ was normalized to 1, and the initial width σ_0 was set to 1.

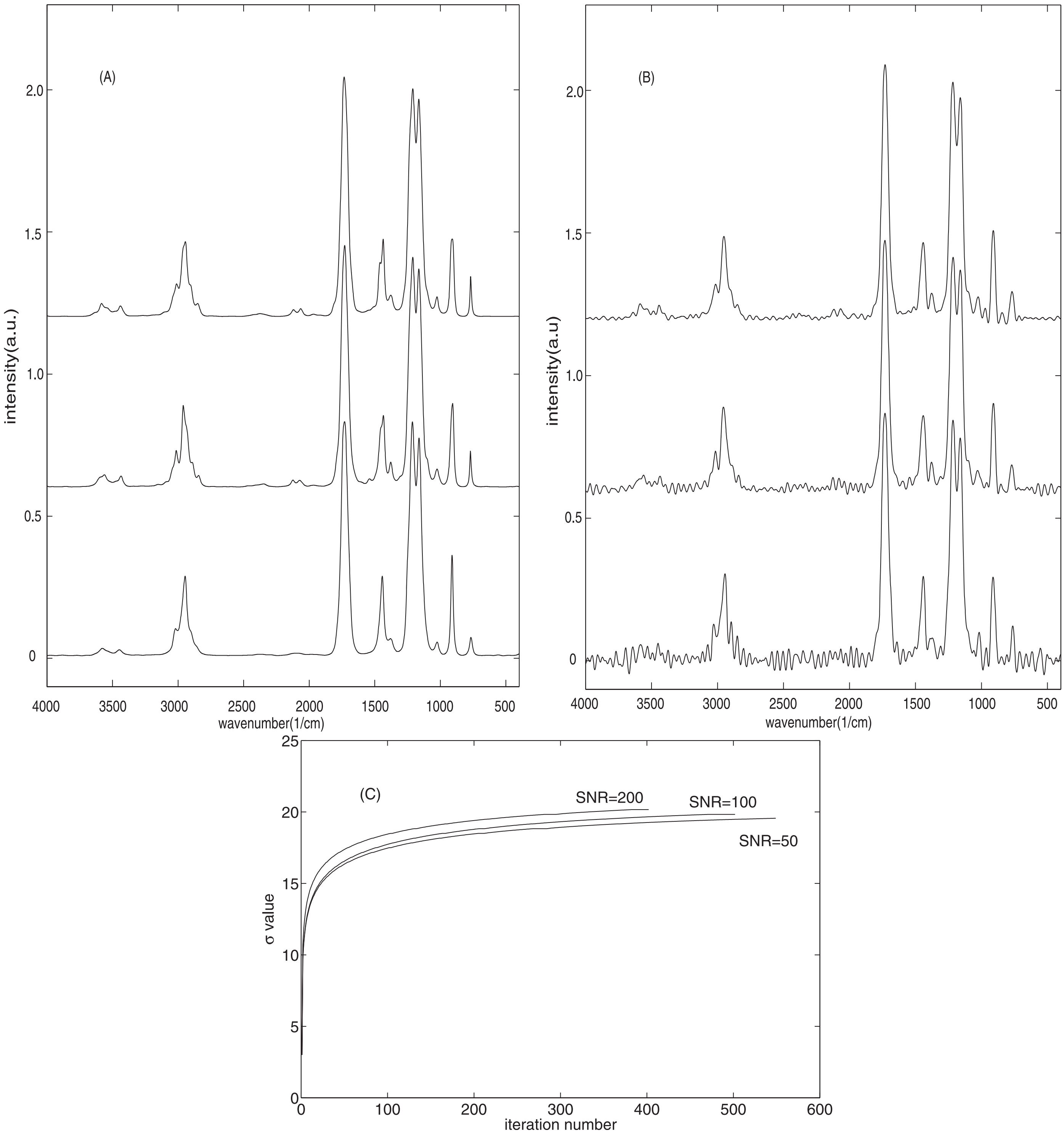

FIG. 5. (**A**) Deconvolution results using MaxEntD on the noisy spectra (from bottom to top SNR of 50, 100, and 200, and NEF of 3.0, 2.5, and 2.0). (**B**) Deconvolution results by SBD on the noisy spectra (from bottom to top SNR of 50, 100, and 200, and $\alpha = 0.05$, $\beta = 300$). (**C**) Convergence curves of the estimated kernel width σ under different SNRs by SBD.

RESULTS AND DISCUSSION

Degraded Spectral Data by Simulation. Following Eq. 1, we simulated degraded spectral data using a computer on the basis of the experimental infrared spectra. Figure 1A presents the IR spectrum of methyl formate ($C_2H_4O_2$) from 4000 to 400 cm^{-1} at 1 cm^{-1} resolution. We convolved it with a Gaussian kernel with a width σ of 20 cm^{-1}, as illustrated in Fig. 1B, to generate the degraded spectrum (Fig. 1C). The degraded spectrum becomes much smoother and less resolved, with the bands becoming wider and lower. For instance, it is difficult to distinguish the peak at 1215 cm^{-1} from that at 1162 cm^{-1}. Afterwards, for investigation of the robustness to noise of the

TABLE II. Figures of merit for the performance of FSD, MaxEntD, and SBD methods in four SNR conditions. For CC and WCC, higher values imply superior performance, while lower RMSE values imply improved performance.

		Method		
Figure of merit	SNR	FSD	MaxEntD	SBD
RMSE	Noise-free	0.0569	0.0158	**0.0144**
	200	0.0569	0.0170	**0.0156**
	100	0.0564	**0.0170**	0.0171
	50	0.0539	**0.0188**	0.0231
CC	Noise-free	0.9841	0.9951	**0.9957**
	200	0.9825	0.9943	**0.9949**
	100	0.9823	**0.9942**	0.9939
	50	0.9821	**0.9934**	0.9890
WCC	Noise-free	0.9889	0.9929	**0.9947**
	200	0.9881	0.9917	**0.9942**
	100	0.9876	0.9914	**0.9941**
	50	0.9869	0.9910	**0.9925**

deconvolution methods, white Gaussian noise was added to the degraded spectrum with three different signal-to-noise ratios (SNRs) of 200, 100, and 50, as shown in Fig. 1D to Fig. 1F.

Comparisons with Other Deconvolution Methods. To investigate the performance of the proposed method and compare it with other methods, we tested three deconvolution methods on the simulated spectra: Fourier self-deconvolution (FSD),[2] maximum entropy deconvolution (MaxEntD),[5] and our semi-blind deconvolution method (SBD). For the SBD method, we tested the performance with and without the adaptive regularization. Because both the FSD and MaxEntD methods need a known PSF in advance, we applied the predefined kernel.

Merits for Performance Evaluation. The performance of the deconvolution methods on the simulated data was evaluated on the basis of the following quantitative merits between the true spectrum f and the recovered spectrum $\hat{f}$:

(a) *Root of mean square error (RMSE):*

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(\hat{f}_i - f_i)^2}{N}} \tag{14}$$

where N is the number of data points of the spectrum, and subscript i denotes the ith data point. RMSE represents the average difference between the two spectra, with a small RMSE corresponding to a good match.

(b) *Pearson correlation coefficient (CC):*

$$CC = \frac{\sum_{i=1}^{N}(f_i - \bar{f})(\hat{f}_i - \bar{\hat{f}})}{\sqrt{\sum_{i=1}^{N}(f_i - \bar{f})^2}\sqrt{\sum_{i=1}^{N}(\hat{f}_i - \bar{\hat{f}})^2}} \tag{15}$$

The horizontal bar denotes the average of all the data points. Pearson's CC represents the average similarity between the trends of the true and the recovered spectra, and the larger values denote the better match, for instance, CC = 1 means a perfect match.

(c) *Griffiths self-weighted correlation coefficient (WCC):*[23]

$$WCC = -\frac{\sum_{i=1}^{N} w_i(f_i - \bar{f})(\hat{f}_i - \bar{\hat{f}})}{\sqrt{\sum_{i=1}^{N} w_i(f_i - \bar{f})^2}\sqrt{\sum_{i=1}^{N} w_i(\hat{f}_i - \bar{\hat{f}})^2}} \tag{16}$$

where w is the weight array, and the horizontal bar denotes the weighted mean. The weight array w was constructed in terms of the intensities of the true spectrum and the differences between the true and recovered spectra.[23] As an improved version of Pearson's CC**,** Griffiths' WCC places emphasis on those intense absorption bands in the spectrum and thus can obtain a more reliable measure of the similarity, which is also consistent with visual comparison. RMSE and Pearson's CC are global metrics that measure the average differences between the two compared spectra, whereas Griffiths' WCC reflects the differences between the local structures of the spectra.

Deconvolution of the Noise-Free Spectrum. First, we applied the three methods on the noise-free degraded spectrum. Figure 2 shows the deconvolution results for the noise-free degraded spectrum in Fig. 1C. For FSD and MaxEntD, we used the accurate Gaussian kernel used in the simulation. For SBD, we started from a Gaussian kernel with an initial width $\sigma_0 = 1$. The width σ converged at about 20 after 500 iterations (see Fig. 3A), and the estimated kernel was plotted in Fig. 3B. The kernel estimated by the SBD method appears to have good similarity to the true kernel In Fig. 2C (SBD method), the peak at 1217 cm^{-1} is distinctly separated from that at 1160 cm^{-1}, while in Figs. 2A (FSD method) and 2B (MaxEntD method), the overlapped bands are separated slightly. Therefore, for the noise-free spectrum, it seems clear that the SBD not only estimates the blur kernel reliably, but also produces a much narrower spectrum with more details than both the FSD and MaxEntD methods.

Also, we tested the MaxEntD method with inaccurate kernels, a narrower and a wider Gaussian kernel, and a Lorentzian kernel. In Figs. 4A and 4B, the predefined Gaussian kernel width is 15 cm^{-1} and 25 cm^{-1}, respectively, while in Fig. 4C a Lorentzian kernel with $\gamma = 15$ cm^{-1} is applied. Compared with the true spectrum (Fig. 1A), the recovery spectrum in Fig. 4A is severely under-deconvoluted, and that in Fig. 4B shows obvious over-deconvolution, whereas that in Fig. 4C differs much from the true one. The test shows that the non-blind deconvolution methods, such as FSD and MaxEntD, cannot achieve reliable results with the inaccurate kernels. On the contrary, the proposed semi-blind method does not suffer these problems.

Furthermore, for the spectra deconvoluted using the MaxEntD and SBD methods, the bands distortions were investigated. In the original spectrum (Fig. 1A), the seven absorbance bands at 2958, 1729, 1434, 1215, 1162, 911, and 768 cm^{-1} are taken as the references. Table I lists these bands distortions and their position, width, height, and area between the deconvoluted (Figs. 2B and 2C) and true (Fig. 1A) spectra. Roots of mean square error (RMSE) of these distortions are calculated. In terms of RMSE, except the position, the distortions in full width at half-maximum (FWHM), height, and area using the SBD method are smaller than those obtained using the MaxEntD method.

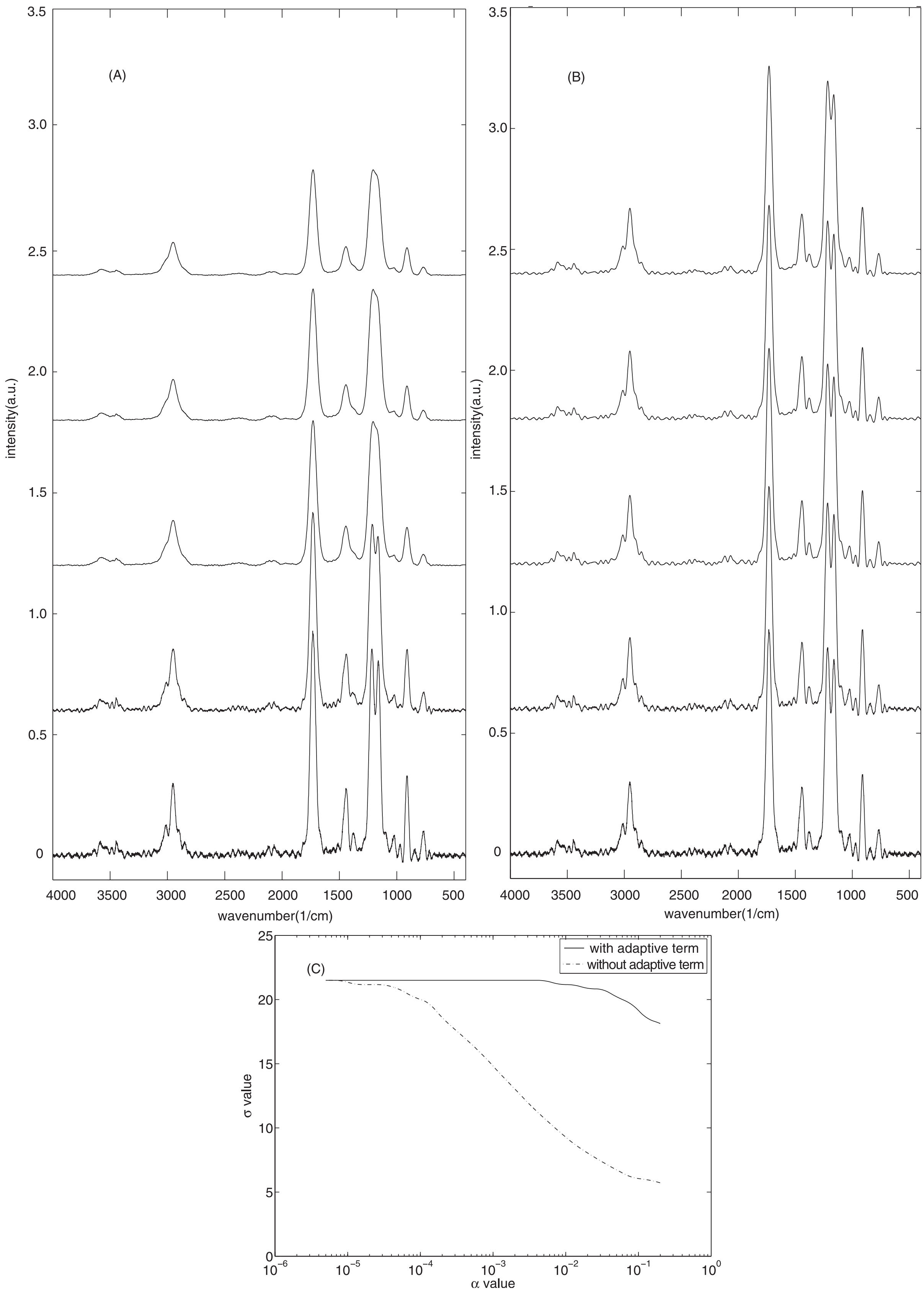

FIG. 6. Deconvolution by SBD method with and without adaptive regularization on the noisy spectrum (SNR = 200). (**A**) Without the adaptive regularization (from bottom to top α of 5×10^{-6}, 5×10^{-4}, 0.05, 0.10, and 0.15). (**B**) With the adaptive regularization (from bottom to top α of 5×10^{-6}, 5×10^{-4}, 0.05, 0.10, and 0.15). (**C**) Estimated kernel width σ as a function of the parameter α.

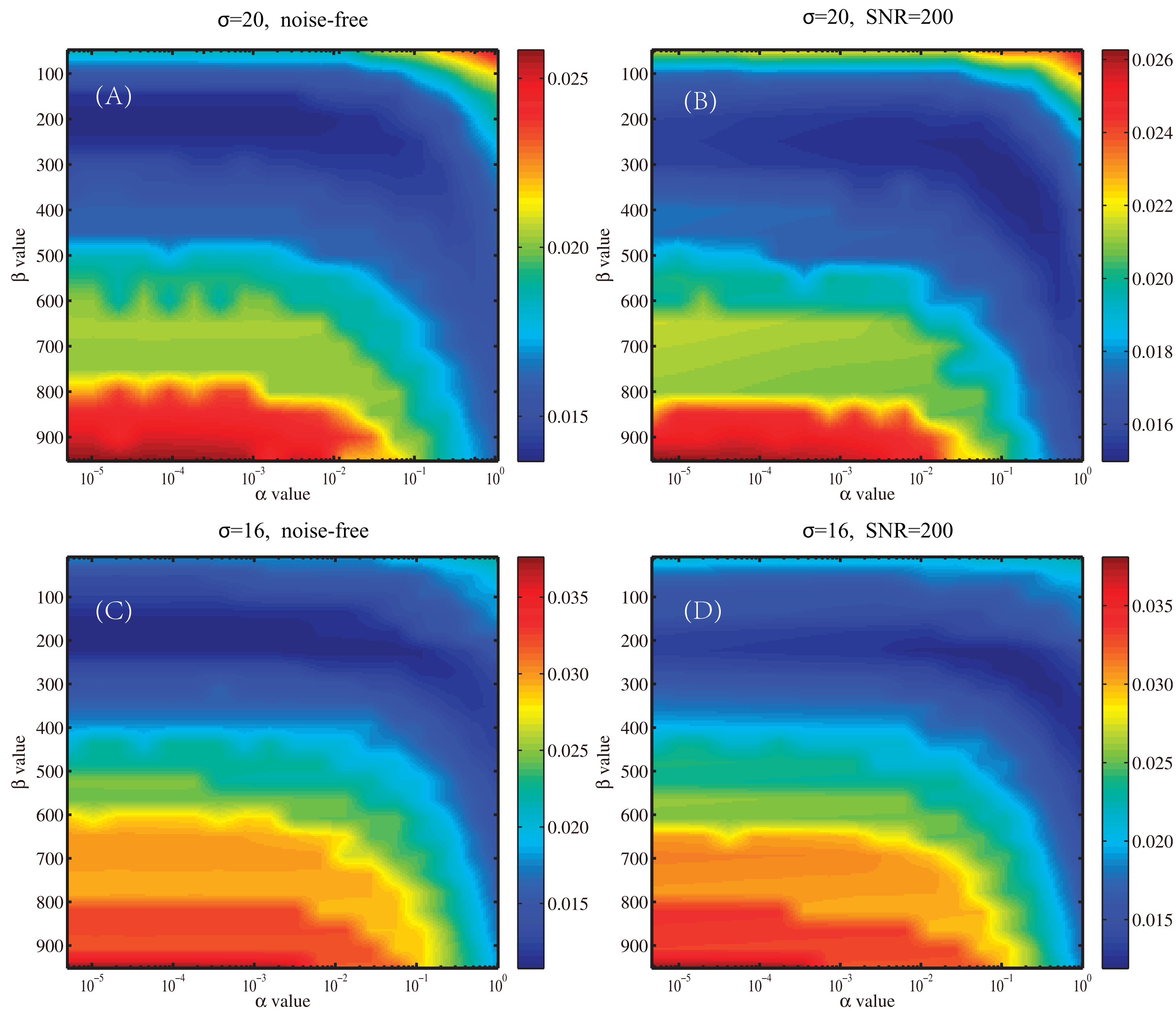

FIG. 7. RMSE versus α and β. (**A**) Width σ = 20, noise-free. (**B**) Width σ = 20, SNR = 200. (**C**) Width σ = 16, noise-free. (**D**) Width σ = 16, SNR = 200. RMSE reaches global minimum when α is between 0.01 and 0.1 and β is between 100 and 300. The results are interpolated to produce smoother plots.

Effect of Noise. We tested the three methods on noisy spectra with SNR of 200, 100, and 50. Figure 5 presents the deconvolution results using MaxEntD and SBD. All the deconvoluted spectra are clearly resolved. For SBD, the deconvoluted spectra appear rather noisy, especially at SNR equal to 50 (see Fig. 5B). On the whole, MaxEntD suppresses the noise better than SBD does. However, for SNR above 50, a more detailed comparison shows that SBD recovers much narrower bands than MaxEntD. For instance, in Fig. 5B, the overlapped bands at 1205 cm^{-1} (see Fig. 1C) are separated more distinctly than those in Fig. 5A. For SNR equal to 50, the two methods present comparable narrowing performance.

Table II shows the resulting RMSE, CC, and WCC for each method on the simulated data with different SNRs. It is clear that MaxEntD and SBD are superior to FSD in all SNR conditions. When the SNR is equal to or higher than 200, SBD achieves a lower RMSE and higher CC than MaxEntD. For SNR below 200, the case is opposite. It shows that MaxEntD suppresses noise better than SBD, especially at low SNRs. Nevertheless, in all SNR conditions SBD gives the highest WCC. As mentioned above, the higher WCC means a better match between the intense bands in the two compared spectra. It means that SBD performs much better than MaxEntD in the recovery of local spectral details, such as the intense bands. Therefore, we can conclude that the proposed SBD method recovers more spectral details at the expense of more residual noise than the MaxEntD method.

Figure 5C shows the convergence curves of the estimated kernel width by SBD from the noisy spectra. It can be seen that, as SNR decreases, the estimated width decreases, and the iterative number for convergence increases. As a result, the recovered spectrum becomes less deconvoluted and with more residual noise (see Fig. 5B, bottom). On the other hand, for better noise suppression, a larger α should be chosen to favor a smoother resolution, which would also result in a narrower kernel, consequently leading to under-deconvolution. This is the reason that the recovery spectrum at SNR = 50 is less deconvoluted than that at SNR = 200. Therefore, for the proposed SBD method, the regularization parameter α should be chosen as a compromise between the narrowing capability and noise suppression, especially at low signal-to-noise ratios.

Effect of the Adaptive Regularization. To demonstrate the effectiveness of the adaptive regularization, we compared the performances of the SBD method with and without the

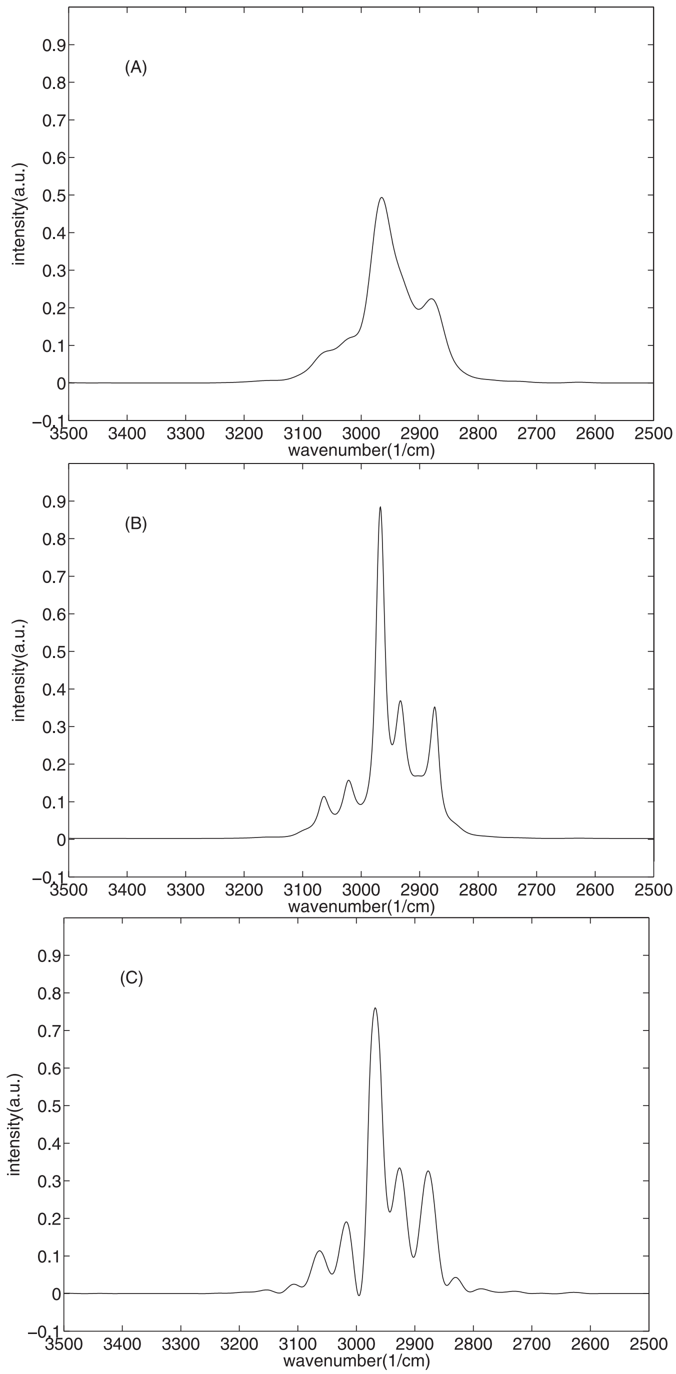

FIG. 8. Deconvolution on experimental infrared spectrum. (**A**) Infrared absorbance spectrum of 1,2-diethylbenzene from 3500 to 2500 cm^{-1} at 1 cm^{-1} resolution. (**B**) MaxEntD deconvolution using the kernel estimated by SBD. (**C**) SBD deconvolution, estimated kernel width $\sigma = 15.8$ ($\alpha = 0.05$, $\beta = 300$).

adaptive regularization. The deconvolution results for the degraded spectrum at SNR of 200, for a parameter α of 5×10^{-6}, 5×10^{-4}, 0.05, 0.10, and 0.15 (from bottom to top) are illustrated in Figs. 6A (without the adaptive regularization) and 6B (with the adaptive regularization). The first effect of the adaptive regularization is an increase of the effective narrowing as well as preservation of details. It is very evident to compare the middle spectra in Figs. 6A and 6B, under the same parameter setting of $\alpha = 0.05$ and $\beta = 300$. There is no deconvoluted effect at all (Fig. 6A, middle). In contrast, due to the adaptive regularization, a rather distinct deconvoluted effect is found from the middle spectrum in Fig. 6B, with sharp bands and fine details. Similar results can be found from the other spectra pairs in Figs. 6A and 6B. The second effect is that the adaptive regularization helps improve the stability of the method with respect to the regularization parameter α. Figure 6C shows the curves of the estimated kernel width σ versus the parameter α at SNR of 200 with and without the adaptive regularization. In the case of without the adaptive regularization, as the parameter α increases from 5×10^{-6} to 0.15, the estimated kernel width σ decreases from 22 to 6. On the contrary, in the case of with the adaptive regularization, the estimated kernel width σ nearly remains at 22 as the parameter α increases from 5×10^{-6} to 5×10^{-2}. It means that without the adaptive regularization the method is unstable with respect to α; in other words, it is difficult to select a proper α to achieve a good estimation for the kernel width. In contrast, benefiting from the adaptive regularization, the resulting kernel width becomes more stable with respect to α. These two positive effects highlight the advantage of the adaptive Tikhonov regularization.

It is noteworthy that the constant k can be determined according to the intensity of the first-order derivative of the IR spectra. Since the intensity of an IR spectrum is normalized to the range [0, 1], in our experiments, the magnitudes of the first-order derivative are in the range [0, 0.05] in most cases. Moreover, the RMSE values change little when the constant k ranges from 0.01 to 0.05. Thus, in our work, for all the simulated and experimental spectra, the constant k was set to be 0.02.

Selecting the Regularization Parameters. The proposed SBD method requires two tuning regularization parameters α and β, which control the smoothness of the recovered spectrum and the kernel, respectively. As α is increased, the recovered spectrum becomes smoother, at the expense of lower fidelity and narrower kernel. Increasing the parameter β would lead to a wider kernel, at the expense of lower fidelity and ringing in the recovered spectrum. Estimation techniques, such as discrepancy principle,[9] generalized cross-validation,[24] and the L-curve method,[25] can be used to determine these regularization parameters automatically. However, the universal applicability of these techniques for different types of spectra and regularization problems has not been effectively validated. In this paper, we conducted a heuristic test on a training dataset to determine the proper ranges for the parameters.

The training dataset was generated on the basis of 30 experimental infrared spectra, which came from the demo infrared spectral library of Bruker Optics Incorporation. The spectral wavenumber ranges from 4000 to 400 at 1 cm^{-1} resolution. First, the spectra were convolved with Gaussian kernels with widths of 16 cm^{-1} and 20 cm^{-1} to generate the blurred spectra. Then, Gaussian noise was added to the blurred spectra with SNRs of 200 and 100. As a result, six training subsets with two blur levels and three SNRs were generated. Each subset contains 30 degraded spectra. The SBD deconvolution tests were performed on the training spectra. Then, the RMSE curve relative to each deconvoluted spectra was plotted as a function of the tuning parameters α and β, respectively. Further, the curve of the average RMSE was obtained by averaging the RMSE values of the 30 spectra in the same

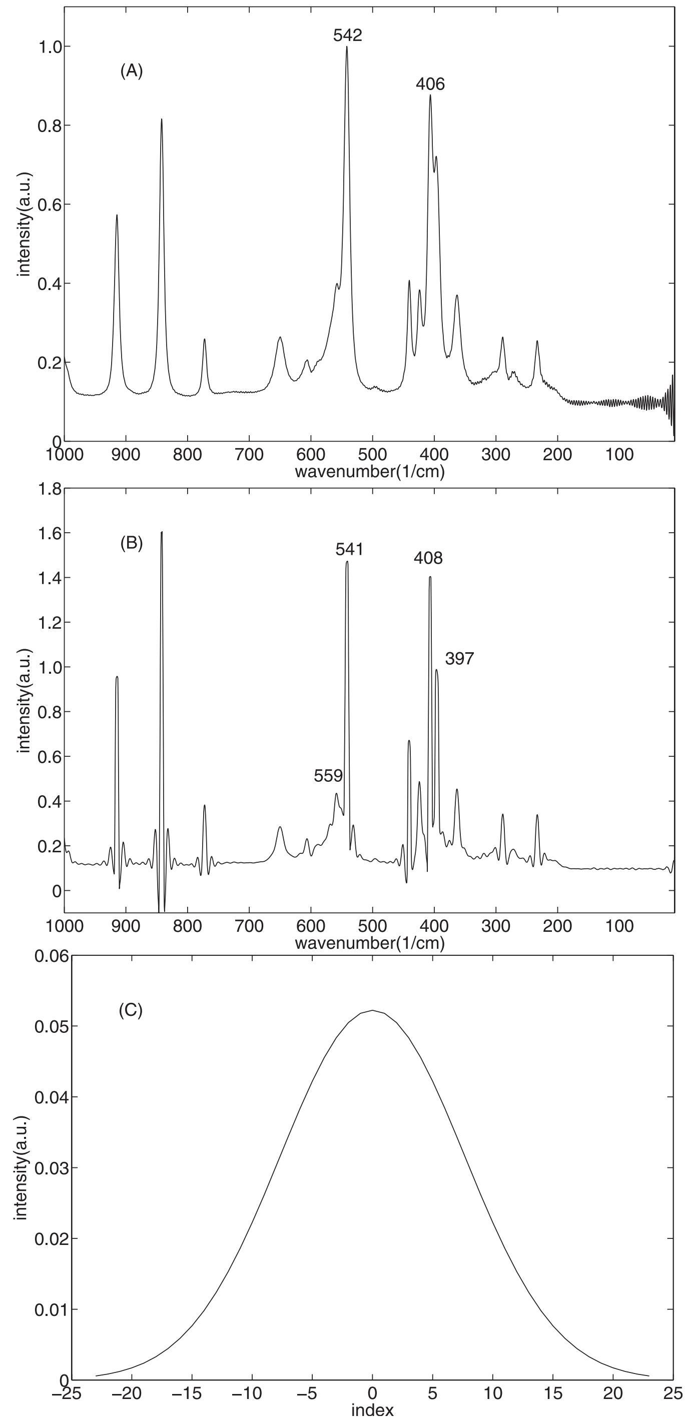

FIG. 9. Deconvolution on experimental Raman spectrum ($\alpha = 0.05$, $\beta = 300$). (**A**) Raman spectrum of (D+)-glucopyranose from 1000 to 1 cm^{-1} at 1 cm^{-1}. (**B**) SBD deconvolution. (**C**) Estimated kernel.

training subset. The parameters leading to the minimization of the RMSE were chosen.

Figure 7 shows the average RMSEs under noise-free and SNR = 200, with kernel widths of 20 cm^{-1} and 16 cm^{-1}. In all conditions, RMSE reaches a global minimum when α is between 0.01 and 0.1 and β is between 100 and 300. These values also lead to desirable kernel widths, and further increase of α will result in a decrease of the estimated kernel width (see Fig. 6C). Therefore, it is advisable for practical infrared spectra to use a value of the parameter α between 0.01 and 0.1 and β between 100 and 300. In this paper, the parameters $\alpha = 0.05$ and $\beta = 300$ were used for the SBD deconvolution.

Deconvolution of Experimental Spectra. Figure 8A shows the infrared absorbance spectrum of 1,2-diethylbenzene from 3500 to 2500 cm^{-1} at 1 cm^{-1} resolution. The spectrum was deconvoluted by MaxEntD using a 15.8 cm^{-1} Gaussian width (equal to the width estimated by SBD) and a NEF of 2.0 (Fig. 8B). Five bands are resolved between 3100 and 2800 cm^{-1}. The SBD method estimated the Gaussian width to be 15.8 cm^{-1}. Six bands are resolved between 3100 and 2800 cm^{-1} (Fig. 8C). It seems that the deconvolution result in Fig. 8C is more resolved than that in Fig. 8B.

The SBD method was also applied to Raman spectra. Figure 9A is the Raman spectrum of (D+)-glucopyranose from 1000 to 1 cm^{-1} at 1 cm^{-1}.[26] The deconvolution spectrum shown in Fig. 9B is far more resolved. The peak at 406 cm^{-1} is split into two peaks at 397 and 408 cm^{-1}, respectively, and the peak at 542 cm^{-1} is split into two peaks at 541 and 559 cm^{-1}, respectively. The estimated kernel is shown in Fig. 9C. The same spectrum has been deconvoluted using the blind method in Yuan et al., where the deconvolution result seems rather noisy.[27] The computing cost was about 33 seconds in Matlab 2008a on a 2.93GHz PC with 2GB memory.

CONCLUSION

We have introduced a semi-blind spectral deconvolution method with adaptive Tikhonov regularization. The method does not require a known PSF in advance but models it as a parametric function in combination with a priori knowledge about the instrumental response characteristics. Moreover, a weighting term is devised in terms of the magnitude of the first derivative of each spectral data to adaptively control the Tikhonov regularization. Consequently, the method can recover a spectrum with the details preserved as well as effectively estimating the blur parameter. Experimental results show the effectiveness of the proposed method. In particular, because the method does not require a known PSF, it has considerable value in practice. It is worth noting that though only the Gaussian kernel is discussed here; any other kernels with parametric forms, such as Lorentzian or Voigt functions, can be applied for the SBD method. We will examine this in a future study.

The introduced method also has some intrinsic limitations. First, it is a linear method, and compared with nonlinear methods, such as maximum entropy-related methods, its noise suppression ability for noisy spectra is still limited. In further study, we plan to introduce some nonlinearity processing to improve the capability of noise suppression. Second, the method needs iteratively performing convolution operations in the signal space, as well as solving linear partial differential equations. It is time consuming and unsuited for real-time applications for the moment. Optimization and acceleration can be further attempted to make the algorithm run fast; for instance, converting the convolution operations in signal space into vector element-wise multiplication in the Fourier domain can reduce computing time. Finally, the proposed method needs tuning of the regularization parameters. Though we have given the typical ranges for the parameters by a heuristic approach, they may not be optimal for practical situations. The optimal parameters could be set manually on a trial and error basis.

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APPENDIX

In this Appendix we derive the Euler–Lagrange equations (Eqs. 7 and 8) that correspond to the object functional (Eq. 5):

$$E(f,\sigma) = \frac{1}{2}\int_L (f\otimes h_\sigma - g)^2\,\mathrm{d}v + \alpha\int_L \rho(|f'|)|f'|^2\,\mathrm{d}v + \beta\int_L |h'_\sigma|^2\,\mathrm{d}v$$

where

$$\rho(|f'|) = \exp\left[-(|f'|/k)^2\right].$$

In the following derivation, let $f(x)$ extend to the whole $\mathbf{R}$ with the Neumann boundary condition, meaning that the function values are uniform along the normal of the spectral boundary. *1) Variation with Respect to f*: The variation of E with respect to f can be expressed as

$$\frac{\delta E}{\delta f} = \frac{\partial}{\partial\lambda}E(f+\lambda\eta)|_{\lambda=0} = \frac{\partial}{\partial\lambda}[E^{\mathrm{I}} + E^{\mathrm{II}}]|_{\lambda=0} = 0$$

where

$$\eta \in C^1(L)$$

with

$$E^{\mathrm{I}} = \frac{1}{2}\int_L [(f+\lambda\eta)\otimes h_\sigma - g]^2\,\mathrm{d}v$$

$$E^{\mathrm{II}} = \alpha\int_L \exp\left[-(|f'+\lambda\eta'|/k)^2\right]|f'+\lambda\eta'|^2\,\mathrm{d}v$$

For the first part

$$\left.\frac{\partial E^{\mathrm{I}}}{\partial\lambda}\right|_{\lambda=0} = \int_L (f\otimes h_\sigma - g)(h_\sigma\otimes\eta)\,\mathrm{d}v$$

By substituting the expression for $h_\sigma \otimes \eta$ and exchanging variables, we obtain

$$\left.\frac{\partial E^{\mathrm{I}}}{\partial\lambda}\right|_{\lambda=0} = \int_L (f\otimes h_\sigma - g)\otimes h_\sigma(-v)\eta(v)\,\mathrm{d}v \qquad (17)$$

For the second part

$$\frac{\partial E^{\mathrm{II}}}{\partial\lambda} = \alpha\int_L \left[\exp\left(-\frac{f'^2 + 2\lambda f'\eta' + \lambda^2\eta'^2}{k^2}\right) \times \left[-\frac{2(f'\eta' + \lambda\eta'^2)}{k^2}\right]|f'+\lambda n'|^2\right] + 2\exp\left(-\frac{f'^2 + 2\lambda f'\eta' + \lambda^2\eta'^2}{k^2}\right)(f'\eta' + \lambda\eta'^2)\,\mathrm{d}v$$

$$\left.\frac{\partial E^{\mathrm{II}}}{\partial\lambda}\right|_{\lambda=0}=\alpha\int_L\left[\exp\left(-\frac{f'^2}{k^2}\right)\left(-\frac{2f'\eta'}{k^2}\right)|f'|^2\right.$$

$$\left.+2\exp\left(-\frac{f'^2}{k^2}\right)f'\eta'\right]\mathrm{d}v$$

$$=\alpha\int_L\exp\left(-\frac{f'^2}{k^2}\right)\left(-\frac{2k^2f'-2f'^3}{k^2}\right)\eta'\,\mathrm{d}v$$

$$=-\alpha\int_L\frac{d}{dv}\left(\exp\left(-\frac{f'^2}{k^2}\right)\left(-\frac{2k^2f'-2f'^3}{k^2}\right)\right)\eta\,\mathrm{d}v$$

$$=-2\alpha\int_L\left(\exp\left(-\frac{f'^2}{k^2}\right)f''\left(\frac{2f'^4}{k^4}-\frac{5f'^2}{k^2}+1\right)\right)\eta\,\mathrm{d}v$$

$$\left.\frac{\partial E^{II}}{\partial\lambda}\right|_{\lambda=0}=-2\alpha\int_L\left[\exp\left(-\frac{f'^2}{k^2}\right)f''\left(\frac{2f'^4}{k^4}-\frac{5f'^2}{k^2}+1\right)\right]\eta\,\mathrm{d}v \tag{18}$$

Adding Eqs. 17 and 18 using the fundamental lemma of calculus of variations, we obtain Eq. 7:

$$\frac{\delta E}{\delta f}=(f\otimes h_\sigma-g)\otimes h_\sigma(-v)$$

$$-2\alpha\exp\left(-\frac{f'^2}{k^2}\right)f''\left(\frac{2f'^4}{k^4}-\frac{5f'^2}{k^2}+1\right)=0$$

2) Differential with Respect to σ: The differential of E with respect to variable σ can be expressed as

$$\frac{\partial E}{\partial\sigma}=\frac{1}{2}\int_L\left[\frac{\partial}{\partial\sigma}(h_\sigma\otimes f-g)^2+\beta\frac{\partial}{\partial\sigma}|h'_\sigma|^2\right]\mathrm{d}v$$

$$=\int_L\left[(h_\sigma\otimes f-g)\left(\frac{\partial h_\sigma}{\partial\sigma}\right)+\beta\frac{\partial}{\partial\sigma}|h'_\sigma|^2\right]\mathrm{d}v=0$$

where

$$\frac{\partial}{\partial\sigma}|h'_\sigma|^2=\frac{v^2}{\pi\sigma^7}\exp\left(-\frac{v^2}{\sigma^2}\right)\left(\frac{v^2}{\sigma^2}-3\right).$$